

OC IT Projects — Chapter 7

Wireless Networks — Quick Review + Quiz

Based on the Chapter 7 training content we studied.

1. Quick Review — طقنللا مهأ

WLAN — Wireless Local Area Network.

Ad Hoc — Devices communicate directly without an Access Point.

Infrastructure — Wireless devices communicate through an Access Point (AP).

IBSS — Independent Basic Service Set; Ad Hoc WLAN with no AP.

BSS — Basic Service Set; one AP. Training example: SOHO.

ESS — Extended Service Set; multiple APs for wider coverage. Training example: college campus.

Mesh Topology — Wireless topology covered in the chapter.

AP Placement — AP location affects coverage; the training discusses 2.4 GHz and 5 GHz placement.

Range Extender — Used to increase wireless coverage/range.

Omnidirectional — Signal radiates in all directions; central AP placement is recommended for this pattern.

Unidirectional / YAGI — Signal is concentrated in one direction. YAGI is used for long-distance/building-to-building links.

MIMO — Multiple Input Multiple Output; multiple transmitting/receiving antennas increase radio-link capacity.

MU-MIMO — Multi-User MIMO; the multiple-antenna concept is used to serve multiple wireless users/terminals.

RFI — Radio Frequency Interference.

CSMA/CA — Carrier Sense Multiple Access / Collision Avoidance; associated with wireless networks.

PSK — Pre-Shared Key; associated with WPA/WPA2 personal usage.

WEP / WPA / WPA2 — Wireless security technologies covered in the training.

802.1X — Network/port-based authentication under Wireless Security.

MAC Filtering — Access control based on MAC addresses.

NAC — Network Admission Control.

Captive Portal — Web-based access/login mechanism after connecting to a wireless network.

Geofencing — Uses geographic location as part of access/control.

Disable SSID Broadcast — Disables SSID broadcasting as a wireless-security measure covered in the training.

2. Must-Memorize Map

- Ad Hoc → No AP
- Infrastructure → AP
- IBSS → No AP → Ad Hoc
- BSS → 1 AP → SOHO
- ESS → Multiple APs → Larger Coverage
- Omnidirectional → All directions → Central AP placement
- Unidirectional → One direction → YAGI → Long distance
- MIMO → Multiple antennas / increased link capacity
- MU-MIMO → Multiple users
- Wireless → CSMA/CA
- PSK → WPA/WPA2 Personal
- 802.1X → Network authentication
- MAC Filtering → MAC-based access control
- NAC → Network Admission Control

3. Chapter 7 — Quiz

25 questions. Answer first, then use the key for revision.

Q1. Which WLAN mode allows devices to communicate without an Access Point?

- A) Infrastructure
- B) Ad Hoc
- C) ESS
- D) BSS

Q2. What does IBSS represent?

- A) A WLAN with multiple APs
- B) A WLAN with one AP
- C) An Ad Hoc WLAN with no AP
- D) A wired network

Q3. A BSS in the training is characterized by:

- A) Multiple APs
- B) No AP
- C) One AP
- D) Two routers

Q4. Which example is associated with an ESS?

- A) College Campus
- B) Single laptop
- C) Small home network with no AP
- D) Point-to-point YAGI link

Q5. What is the main difference between BSS and ESS?

- A) BSS uses cables while ESS is wireless
- B) BSS has one AP; ESS has multiple APs
- C) BSS has no AP; ESS has one AP
- D) BSS uses MIMO; ESS uses MU-MIMO

Q6. Which antenna radiates its signal power in all directions?

- A) YAGI
- B) Unidirectional
- C) Omnidirectional
- D) Mesh antenna

Q7. Where does the training recommend placing an AP when using an omnidirectional antenna?

- A) At the edge of the room
- B) In the center of the room
- C) Outside the building
- D) Next to the router only

Q8. A company wants to connect two buildings over a long distance using focused wireless signals. Which is most appropriate?

- A) Omnidirectional antenna
- B) YAGI antenna
- C) Range extender
- D) BSS

Q9. What does “Uni” in Unidirectional indicate?

- A) One direction
- B) One AP
- C) One frequency
- D) One user

Q10. What is the purpose of a Wireless Range Extender?

- A) Encrypt wireless traffic
- B) Increase wireless coverage/range
- C) Replace the IP address
- D) Create a collision domain

Q11. What does MIMO use to increase the capacity of a radio link?

- A) Multiple transmission and receiving antennas
- B) Multiple IP addresses
- C) Multiple AP passwords
- D) Multiple SSIDs only

Q12. The key idea behind MU-MIMO is communication involving:

- A) Multiple cables
- B) Multiple switches
- C) Multiple users/wireless terminals
- D) Multiple passwords

Q13. RFI stands for:

- A) Radio Frequency Interface
- B) Radio Frequency Interference
- C) Remote Frequency Internet
- D) Router Frequency Isolation

Q14. Which access method is associated with wireless networks in this training?

- A) CSMA/CD
- B) CSMA/CA
- C) CDMA/CD
- D) CSMA/CS

Q15. Which statement correctly compares Ethernet and Wireless access methods?

- A) Ethernet → CSMA/CA, Wireless → CSMA/CD
- B) Both always use CSMA/CD
- C) Ethernet → CSMA/CD, Wireless → CSMA/CA
- D) Both always use CSMA/CA

Q16. Which security mechanism is associated with WPA/WPA2 personal usage?

- A) PSK
- B) TDR
- C) DHCP
- D) NAT

Q17. Which of the following is listed under Wireless Security?

- A) FTP
- B) 802.1X
- C) DNS
- D) SMTP

Q18. What does NAC stand for?

- A) Network Access Channel
- B) Network Authentication Cable
- C) Network Admission Control
- D) Network Address Configuration

Q19. Which technology can present a web page for users after they connect to a Wi-Fi network?

- A) Captive Portal
- B) MIMO
- C) YAGI
- D) BSS

Q20. What is MAC Address Filtering primarily based on?

- A) User's physical location
- B) MAC addresses
- C) Frequency channels
- D) Antenna direction

Q21. Scenario: A small office has one AP connected to the network. Which service set best describes it?

- A) IBSS
- B) ESS
- C) BSS
- D) Mesh

Q22. Scenario: A university deploys multiple APs across a large campus. Which service set is most appropriate?

- A) BSS
- B) ESS
- C) IBSS
- D) Ad Hoc

Q23. Scenario: Two laptops communicate wirelessly directly with no Access Point. Which configuration matches?

- A) Infrastructure
- B) ESS
- C) BSS
- D) Ad Hoc / IBSS

Q24. Scenario: An antenna concentrates power toward one direction for a long-distance building-to-building link. Which choice is best?

- A) Omnidirectional
- B) YAGI / Unidirectional
- C) Range Extender
- D) BSS

Q25. Scenario: One technology improves a radio link using multiple antennas, while another is designed to serve multiple wireless users. Which pairing is correct?

- A) MIMO → multiple users; MU-MIMO → one antenna
- B) MIMO → multiple antennas/link capacity; MU-MIMO → multiple users
- C) MIMO → security; MU-MIMO → authentication
- D) MIMO → AP placement; MU-MIMO → RFI

4. Answer Key

1-B | 2-C | 3-C | 4-A | 5-B

6-C | 7-B | 8-B | 9-A | 10-B

11-A | 12-C | 13-B | 14-B | 15-C

16-A | 17-B | 18-C | 19-A | 20-B

21-C | 22-B | 23-D | 24-B | 25-B

5. Session Result & Points to Review

22/25 — 88%

Review: RFI = Radio Frequency Interference; BSS = one AP (SOHO); MIMO = multiple antennas/link capacity; MU-MIMO = multiple users.